

TRIBHUVAN UNIVERSITY

Institute of Science and Technology

A CASE STUDY REPORT

ON

NEPAL'S POSITION IN THE UNITED NATIONS E-GOVERNMENT DEVELOPMENT INDEX (EGDI)

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A CASE STUDY REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR IN SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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EXECUTIVE SUMMARY

This case study report presents a comprehensive technical and policy appraisal of Nepal's position in the United Nations E-Government Development Index (EGDI), the principal internationally recognized composite instrument through which the United Nations Department of Economic and Social Affairs (UN DESA) evaluates the maturity of digital governance across all 193 member states. In the thirteenth edition of the biennial E-Government Survey, published in September 2024, Nepal recorded an EGDI score of 0.5781 and secured a global rank of 119 out of 193 countries, thereby formally entering the High EGDI (HEGDI) rating class for the first time in its national history. This represents an advance of six positions and a score gain of 0.0664 over the 2022 survey cycle, in which Nepal had ranked 125th with a score of 0.5117.

The report's central technical finding is that Nepal's climb into the High EGDI group has been driven overwhelmingly by an asymmetric surge in the Telecommunication Infrastructure Index (TII), which reached 0.7653 in 2024, a value that now substantially exceeds the country's Human Capital Index (HCI) of 0.5210 and its Online Service Index (OSI) of 0.4481. This divergence, herein termed the OSI TII asymmetry, indicates that Nepal's improved standing is predominantly an artifact of mobile and broadband network expansion (leapfrogging via 4G penetration and the Rural Telecommunications Development Fund optical backbone) rather than a genuine transformation in the transactional depth, interoperability, or citizen usability of government digital services. Compounding this concern, Nepal's supplementary E-Participation Index (EPI) stood at only 0.2192 in 2024, corresponding to a global rank of 152, indicating that despite improved connectivity, mechanisms for e-information sharing, e-consultation, and e-decision making between the state and its citizens remain severely underdeveloped.

Regionally, Nepal continues to trail every South Asian peer with the sole exceptions of Pakistan and Afghanistan, positioned behind the Maldives (rank 94), India (rank 97), Sri Lanka (rank 98), Bangladesh (rank 100), and Bhutan (rank 103). This regional benchmarking exercise demonstrates that landlocked developing country (LLDC) status, combined with federal restructuring complexities, chronic institutional data fragmentation, and constrained civil service technical capacity, continues to impede Nepal's transition toward the Very High EGDI (VHEGDI) category, which requires a composite score exceeding 0.75.

At the domestic policy level, the report undertakes a detailed audit of the Digital Nepal Framework (DNF) 2019, the Electronic Transactions Act 2063, the National Cyber Security Policy 2080, and flagship citizen facing platforms including the Nagarik App, the National Identity Card system administered by the Department of National ID and Civil Registration (DoNIDCR), the electronic Government Procurement (e-GP) system operated by the Public Procurement Monitoring Office (PPMO), the Integrated Tax System of the Inland Revenue Department (IRD), the Electronic Driving License and Vehicle Registration System (EDLRS) of the Department of Transport Management (DoTM), the Land Records Information Management System (LRIMS), and the SUTRA municipal financial management platform. The audit finds that while the breadth of platform proliferation is considerable, transactional maturity, cross agency interoperability under the Nepal Government Enterprise Architecture (NGEA), and adherence to the Nepal Government Interoperability Framework (NeGIF) remain structurally weak.

The report concludes with a phased strategic roadmap and a set of actionable, stakeholder specific recommendations directed at federal policy makers, infrastructure operators, implementing ministries, subnational governments, and the academic and domestic ICT industry ecosystem. The unifying strategic imperative identified throughout this report is the necessity of transitioning Nepal's digital governance model from one of portal proliferation and infrastructure accumulation toward one of integrated, secure, and citizen centric transactional governance, without which further ascent toward the Very High EGDI category will remain structurally constrained irrespective of continued telecommunications investment.

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CHAPTER 1: INTRODUCTION

Electronic governance, commonly abbreviated as e-government, refers to the application of information and communication technologies (ICT) by public sector institutions to improve the efficiency, transparency, accessibility, and accountability of governmental functions and citizen services. As a field of academic and applied inquiry, e-government sits at the intersection of computer science, public administration, and development economics, and its theoretical foundations rest upon the concept of public value creation, a framework under which the legitimacy of technological investment in the public sector is judged not merely by systems uptime or infrastructural sophistication, but by the tangible improvements delivered to citizens in terms of service quality, participatory governance, and institutional trust.

Since the early 2000s, the United Nations has positioned itself as the principal global authority in benchmarking the digital maturity of national governments through its biennial E-Government Survey, first published in 2001 and refined through thirteen subsequent editions culminating in the 2024 release. The resulting E-Government Development Index (EGDI) has become the most widely cited comparative metric of digital governance capacity worldwide, informing national ICT policy formulation, donor prioritization, and academic research across more than two decades of longitudinal data collection.

For Nepal, a landlocked, federally restructured, lower middle income country undergoing a prolonged and still incomplete transition from a centralized unitary administrative structure to a three tier federal system comprising 753 local governments, 7 provinces, and the federal government, the EGDI carries particular analytical significance. It functions simultaneously as an external validation instrument, a domestic policy accountability mechanism, and a diagnostic tool capable of revealing structural asymmetries between different pillars of digital governance capacity that might otherwise remain obscured within aggregate national development narratives.

1.1 The United Nations EGDI Conceptual Model

The EGDI is constructed as an equal weighted composite index derived from three normalized sub indices, namely the Online Service Index (OSI), the Telecommunication Infrastructure Index (TII), and the Human Capital Index (HCI). The conceptual architecture of the index reflects a deliberate methodological choice by UN DESA to capture digital governance not as a unidimensional phenomenon reducible to the sophistication of a national government portal, but as a multidimensional construct requiring the simultaneous presence of transactional online services, physical and network telecommunications infrastructure, and an educated citizenry capable of engaging meaningfully with digital public services.

The institutional custodian of the survey is the Division for Public Institutions and Digital Government within UN DESA, which coordinates data collection through a combination of an Online Services Questionnaire administered to expert evaluators who assess national government websites and portals, secondary telecommunications data sourced from the International Telecommunication Union (ITU), and human capital indicators sourced from UNESCO and the United Nations Population Division. This triangulated data collection methodology lends the EGDI a degree of empirical rigor and cross national comparability that few alternative digital governance metrics possess, although, as later chapters of this report will demonstrate, the index's equal weighting scheme can also produce compensatory effects wherein strong performance in one sub index (most notably telecommunications infrastructure) can mask persistent weaknesses in others (most notably online transactional services), a phenomenon of direct relevance to the Nepali case.

1.2 Scope of the Report

This report confines its analytical scope to publicly documented and civilian facing dimensions of e-governance performance across the federal, provincial, and municipal tiers of the Nepali state. Specifically, the report examines national portals, citizen facing transactional platforms, telecommunications infrastructure investment programs, human capital and digital literacy indicators, and the institutional and legal frameworks that govern digital public service delivery. Matters pertaining to classified national security networks, defense sector digital systems, or

intelligence infrastructure fall explicitly outside the scope of this report and are not addressed at any point in the subsequent chapters.

The temporal scope of the report spans the complete longitudinal record of UN E-Government Survey publications, from the inaugural 2003 edition through the most recent 2024 edition, with forward looking policy recommendations extending toward the 2026 survey cycle and beyond, toward the strategic horizon of Nepal's Fifteenth and Sixteenth periodic development plans.

1.3 Methodology

The research methodology adopted for this case study is a longitudinal, desk based, secondary data analysis. Primary sources of quantitative data include the full series of UN DESA E-Government Survey publications and their accompanying technical appendices and statistical annexes, supplemented by data from the International Telecommunication Union's World Telecommunication and ICT Indicators database, World Bank open data indicators pertaining to gross national income, internet penetration, and educational attainment, and official disclosures published by Nepal's Ministry of Communications and Information Technology (MoCIT), the National Information Technology Center (NITC), and the Nepal Telecommunications Authority (NTA).

The analytical method employed combines descriptive time series analysis of Nepal's EGDI trajectory, comparative cross sectional benchmarking against South Asian regional peers, and qualitative policy document analysis of the Digital Nepal Framework 2019, the Electronic Transactions Act 2063, and the National Cyber Security Policy 2080. Where sub index values for historical survey years were not directly disaggregated in secondary reporting, the Human Capital Index component was computed by the researcher using the official equal weighted aggregation formula (EGDI equals the arithmetic mean of the normalized OSI, TII, and HCI values) applied in reverse against the published composite EGDI and the independently reported OSI and TII values for that year; such computed figures are clearly identified as derived values in the relevant tables.

1.4 Research Questions and Problem Formulation

The report is structured around the following central research questions, each addressed substantively in the analytical chapters that follow:

1. Why has Nepal's Telecommunication Infrastructure Index advanced at a substantially faster rate than its Online Service Index and E-Participation Index across successive survey cycles, and what structural, institutional, and investment factors account for this divergence.
2. To what extent do persistent data silos among federal ministries, provincial agencies, and local governments constrain Nepal's ability to deliver integrated, interoperable, transactional digital public services.
3. What are the principal causes of Nepal's chronically weak E-Participation Index performance, and what institutional reforms would be required to strengthen citizen facing consultative and participatory digital mechanisms.
4. What operational, technical, and human resource factors contribute to service downtime, concurrency failures, and reliability deficits observed across major Nepali citizen platforms, and what remediation pathway would allow Nepal to advance from the High EGDI category toward the Very High EGDI category.

CHAPTER 2: METHODOLOGY AND MATHEMATICAL FRAMEWORK OF THE UN EGDI

This chapter formalizes the mathematical and technical architecture underlying the UN EGDI framework, providing the quantitative foundation against which Nepal's longitudinal and comparative performance is assessed throughout the remainder of this report.

2.1 The Composite EGDI Equation and Normalization Procedures

The Composite EGDI for a given member state i in survey year t is formally defined as the equal weighted arithmetic mean of three normalized sub index scores, expressed mathematically as follows.

EGDI(i,t) = one third multiplied by the sum of Normalized OSI(i,t), Normalized TII(i,t), and Normalized HCI(i,t)

Each of the three constituent sub indices is itself a composite of multiple underlying indicators, each of which is first standardized using a min max normalization procedure (in earlier survey editions) or a hybrid z score and min max transformation (in more recent editions, adopted from the 2018 survey onward to reduce sensitivity to extreme outlier values among high income small states). The normalization transformation for a given indicator value X for country i is expressed as follows.

Normalized X(i) equals the quantity X(i) minus the minimum observed value of X across all countries, divided by the quantity representing the maximum observed value of X minus the minimum observed value of X across all countries.

This procedure rescales every underlying indicator onto a common zero to one interval prior to aggregation into its parent sub index, and each parent sub index is subsequently rescaled onto the same zero to one interval prior to entering the final composite EGDI calculation. The resulting composite EGDI score therefore always falls within the closed interval of zero to one, permitting UN DESA to classify member states into four ordinal rating classes: Low EGDI (scores from 0.00 to 0.25), Middle EGDI (scores from 0.25 to 0.50), High EGDI (scores from 0.50 to 0.75), and Very

High EGDI (scores from 0.75 to 1.00). Nepal's 2024 score of 0.5781 places it firmly within the High EGDI band, in the second of three sub bands (H2) that UN DESA uses for more granular internal classification within the High EGDI category.

2.2 Online Service Index (OSI) Evaluation Criteria

The Online Service Index evaluates the scope, quality, and transactional depth of a country's national government online presence through an Online Services Questionnaire comprising more than one hundred and forty distinct evaluative questions in the 2024 edition, organized around five conceptual sub dimensions: institutional framework (the presence of governance structures, digital strategy documents, and legal provisions supporting digital service delivery), service provision (the depth to which services progress from static informational presence through interactive engagement to full end to end transactional completion), content provision (the availability, currency, and multilingual accessibility of government information), technology (the deployment of mobile compatibility, accessibility features for persons with disabilities, open data formats, and application programming interfaces), and e-participation (addressed independently as a supplementary index but also feeding into the OSI's institutional assessment). Nepal's 2024 OSI value of 0.4481 places the country within UN DESA's middle OSI performance band, alongside comparator states such as Algeria, Botswana, and Cambodia, reflecting persistent limitations in transactional service maturity relative to the country's underlying network infrastructure.

2.3 Telecommunication Infrastructure Index (TII) Indicators

The Telecommunication Infrastructure Index is constructed from five underlying ITU sourced indicators measured on a per capita or per population basis: the number of fixed telephone subscriptions, the number of mobile cellular telephone subscriptions, the number of active mobile broadband subscriptions, the number of fixed (wired) broadband subscriptions, and the estimated proportion of individuals using the internet. Nepal's TII value of 0.7653 in 2024, a dramatic increase from 0.4691 in 2020 and 0.2413 in 2018, reflects the compounding effects of rapid 4G network densification by the two dominant mobile network operators, Nepal Telecom and Ncell, alongside substantial growth in mobile broadband subscriptions driven by falling smartphone costs and expanded 4G tower coverage across the hill and mountain regions following the completion

of major segments of the Rural Telecommunications Development Fund (RTDF) supported optical fiber backbone.

2.4 Human Capital Index (HCI) Indicators

The Human Capital Index is constructed from four UNESCO sourced educational indicators: the adult literacy rate, the gross enrollment ratio (combined for primary, secondary, and tertiary education), the expected years of schooling for a child entering the education system, and the mean years of schooling actually completed by adults aged twenty five and above. Nepal's HCI value of 0.5210 in 2024 reflects continued gradual improvement in national literacy (now estimated above 76 percent among the adult population) and rising tertiary enrollment, but remains constrained by disparities in educational quality and access between urban centers such as Kathmandu and remote hill and Himalayan districts, together with a persistent gap between general literacy and the specific digital and ICT literacy competencies required for meaningful engagement with transactional e-government platforms.

2.5 Supplementary Framework: E-Participation Index (EPI)

The E-Participation Index is a supplementary index computed independently of the three core EGDI sub indices, though it draws upon a subset of OSI questionnaire items relating specifically to citizen state digital interaction. The EPI is organized around three progressive stages of digital participation: e-information (the extent to which a government proactively publishes information relevant to public policy, budgets, and legislation online), e-consultation (the extent to which a government provides functional online mechanisms for citizens to comment upon or deliberate over draft policies, legislation, and budgets), and e-decision making (the extent to which a government demonstrably incorporates the outcomes of online citizen consultation into final policy and administrative decisions, and communicates this incorporation back to the public). Nepal's 2024 EPI value of 0.2192, corresponding to a global rank of 152 out of 193, indicates severe underdevelopment across all three stages, but particularly within e-consultation and e-decision making, where functional feedback loops between citizen input and administrative or legislative outcomes remain almost entirely absent from major government portals.

Table 1: Mathematical Structure, Indicators, and Weighting Schema of the UN EGDI Framework

Component	Weight in Composite EGDI	Number of Underlying Indicators (2024)	Primary Data Source	Nepal 2024 Value
Online Service Index (OSI)	One Third (Equal Weight)	Approximately 140 questionnaire items across 5 sub dimensions	UN DESA Online Services Questionnaire	0.4481
Telecommunication Infrastructure Index (TII)	One Third (Equal Weight)	5 indicators	International Telecommunication Union (ITU)	0.7653
Human Capital Index (HCI)	One Third (Equal Weight)	4 indicators	UNESCO Institute for Statistics; UN Population Division	0.5210
E-Participation Index (EPI)	Supplementary (Not Weighted into Composite EGDI)	Approximately 30 questionnaire items across 3 stages	UN DESA Online Services Questionnaire	0.2192

CHAPTER 3: OBJECTIVES OF THE STUDY

Primary Objectives

1. To conduct a rigorous, empirically grounded assessment of Nepal's position, trajectory, and structural composition within the United Nations E-Government Development Index across all available survey cycles from 2003 through 2024.
2. To identify and analytically explain the asymmetric growth pattern between Nepal's Telecommunication Infrastructure Index, Human Capital Index, and Online Service Index, and to determine the structural and institutional factors underlying this divergence.
3. To benchmark Nepal's e-government performance against South Asian regional peers, including India, the Maldives, Sri Lanka, Bangladesh, Bhutan, Pakistan, and Afghanistan, across all EGDI sub indices and the supplementary E-Participation Index.
4. To evaluate the alignment between Nepal's principal domestic digital governance policy instruments, most notably the Digital Nepal Framework 2019, and the technical dimensions measured by the UN EGDI methodology.

Secondary Objectives

1. To audit the operational status, technology stack maturity, and transactional depth of Nepal's principal citizen facing digital platforms, including the Nagarik App, the National Identity Card system, e-Passport issuance, the e-Government Procurement system, the Integrated Tax System, and land records digitization.
2. To assess Nepal's national cybersecurity posture, incident response capacity, and data protection framework in relation to the risks inherent in expanding transactional digital government infrastructure.
3. To formulate a phased, stakeholder specific set of policy and engineering recommendations capable of guiding Nepal's transition from the High EGDI category toward the Very High EGDI category within a realistic medium term planning horizon.

4. To contribute a structured academic case study to the Nepali computer science and public administration literature addressing the intersection of national ICT policy, telecommunications infrastructure investment, and international digital governance benchmarking.

CHAPTER 4: LONGITUDINAL EVALUATION OF NEPAL'S EGDI TRAJECTORY (2003 TO 2026)

4.1 Historical Performance Analysis

Nepal's EGDI trajectory across the thirteen survey editions published between 2003 and 2024 exhibits a distinctive three phase pattern: an initial period of stagnation and gradual decline between 2003 and 2014, a period of post 2015 institutional recovery coinciding with the promulgation of the 2015 Constitution and the launch of federal restructuring, and a more recent phase of accelerated post pandemic infrastructural gain between 2020 and 2024.

In the inaugural 2003 survey, Nepal registered an EGDI score of 0.2684 and a global rank of 130 out of the then smaller cohort of assessed member states. Rather than improving, Nepal's relative position deteriorated steadily across the subsequent decade, falling to rank 150 in 2008, rank 153 in 2010, rank 164 in 2012, and reaching its historical nadir in the 2014 survey at rank 165 with a score of only 0.2344, the lowest EGDI value Nepal has recorded across the entire survey series. This period of stagnation is generally attributed to the decade long Maoist insurgency and its aftermath, chronic political instability with frequent changes of government, and minimal sustained public investment in telecommunications infrastructure or digital public administration reform during this period.

A pronounced recovery began with the 2016 survey, in which Nepal's score rose to 0.3458 (rank 135), followed by a substantial leap in the 2018 survey to 0.4748 (rank 117), Nepal's strongest relative improvement in the survey's history at that time, driven in large part by a temporary surge in the Online Service Index to 0.6875 following the launch of several new government portals in the aftermath of the 2015 earthquake reconstruction effort and the promulgation of the new federal Constitution. The 2020 survey recorded a decline to rank 132 (score 0.4699), a regression widely attributed to the operational disruptions, budget reallocations, and service delivery interruptions experienced across Nepali public institutions during the COVID nineteen pandemic, notwithstanding a continued rise in the underlying Telecommunication Infrastructure Index during the same period. Nepal recovered to rank 125 in the 2022 survey (score 0.5117) and advanced further to its historic best position of rank 119 in the 2024 survey, with a score of 0.5781, formally

crossing the 0.50 threshold that separates the Middle EGDI and High EGDI rating classes for the first time in the country's survey history.

4.2 Sub Component Deconstruction

TII Evolution: Leapfrogging via Mobile Telecommunications

Nepal's Telecommunication Infrastructure Index exhibits by far the most dramatic longitudinal improvement of the three EGDI sub indices, rising from a value of only 0.0064 in 2003, reflecting near total absence of fixed line and early mobile infrastructure, to 0.7653 in 2024, an increase of more than one hundred and nineteen fold across two decades. This trajectory reflects a classic telecommunications leapfrogging pattern, in which Nepal bypassed large scale fixed line network investment entirely in favor of mobile cellular and, subsequently, mobile broadband infrastructure. The steepest gains occurred between 2018 and 2024, coinciding with nationwide 4G license rollouts by Nepal Telecom and Ncell, substantial completion of Rural Telecommunications Development Fund financed optical fiber corridors connecting district headquarters, and the operationalization of the Government Integrated Data Center (GIDC) at Singha Durbar alongside the disaster recovery center at Hetauda, which together provide the sovereign hosting infrastructure underpinning national digital platforms.

HCI Progression: Educational Indicators and Civil Service ICT Skill Ceilings

Nepal's Human Capital Index has progressed more modestly, from an estimated 0.48 in 2003 to 0.5210 in 2024, reflecting the combined effects of rising national literacy, expanding secondary and tertiary gross enrollment ratios, and demographic shifts including sustained rural to urban migration and international labor outmigration that have altered the composition of the resident working age population. However, aggregate national literacy and enrollment statistics substantially overstate the specific ICT and digital literacy competencies present within the civil service workforce responsible for operating and maintaining e-government platforms. Persistent wage compression within government technical cadres, combined with an absence of a dedicated government wide information technology service cadre comparable to those established in India or Sri Lanka, has constrained the depth of institutional digital capacity even as broader educational indicators have improved.

OSI Stagnation: The Disconnect Between Policy Blueprints and Transactional Digital Maturity

In marked contrast to the TII trajectory, Nepal's Online Service Index has followed a volatile, largely stagnant path, moving from 0.3188 in 2003 to a temporary peak of 0.6875 in 2018 before declining to 0.4000 in 2020 and settling at 0.4481 in 2024, a value only marginally higher than the country's 2005 level of 0.4000. This near two decade long plateau in transactional online service maturity, occurring in parallel with more than a one hundred fold increase in underlying telecommunications infrastructure, constitutes the single most significant structural finding of this report and is examined in depth in Chapter 7.

4.3 The E-Participation Conundrum

Nepal's E-Participation Index performance across the survey series has been markedly more volatile than any other sub component, rising from a low base of 0.1379 in 2003 to a peak of 0.7809 in 2018 (rank 55, Nepal's best ever EPI performance, driven by temporary e-consultation features associated with post earthquake reconstruction planning portals) before collapsing to 0.3690 in 2020, 0.2386 in 2022, and 0.2192 in 2024 (rank 152). This collapse indicates that Nepal's 2018 EPI peak reflected transient, project specific consultation platforms rather than institutionalized, durable participatory digital infrastructure, and that no sustained mechanism for structured e-consultation or e-decision making has since been embedded within core government portals at the federal, provincial, or municipal level.

Table 2: Longitudinal EGDI, OSI, TII, and HCI Data for Nepal (2003 to 2024)

Survey Year	Global Rank	EGDI Score	OSI	TII	HCI (Derived where noted)
2003	130	0.2684	0.3188	0.0064	0.4800*
2004	132	0.2807	0.3359	0.0063	0.5000*
2005	126	0.3021	0.4000	0.0063	0.5000*
2008	150	0.2725	0.2876	0.0119	0.5180*
2010	153	0.2568	0.1683	0.0227	0.5794*
2012	164	0.2664	0.2876	0.0597	0.4521*
2014	165	0.2344	0.1575	0.1684	0.3774*
2016	135	0.3458	0.3986	0.1675	0.4714*

Survey Year	Global Rank	EGDI Score	OSI	TII	HCI (Derived where noted)
2018	117	0.4748	0.6875	0.2413	0.4956*
2020	132	0.4699	0.4000	0.4691	0.5406*
2022	125	0.5117	0.4592	0.5123	0.5636*
2024	119	0.5781	0.4481	0.7653	0.5210

** Human Capital Index values for survey years prior to 2024 were computed by the researcher by inverting the equal weighted composite EGDI formula against the officially published EGDI, OSI, and TII values for that year, since disaggregated HCI figures were not consistently available in secondary reporting for all historical editions.*

Table 3: Longitudinal E-Participation Index (EPI) Performance and Global Ranks for Nepal

Survey Year	EPI Value	Global Rank
2003	0.1379	61
2004	0.0656	75
2005	0.0794	73
2008	0.0227	152
2010	0.0571	127
2012	0.0263	134
2014	0.2941	110
2016	0.5085	89
2018	0.7809	55
2020	0.3690	137
2022	0.2386	143
2024	0.2192	152

CHAPTER 5: REGIONAL BENCHMARKING: NEPAL IN SOUTH ASIA

5.1 Comparative Regional Landscape

Within the South Asian sub region as classified by UN DESA, Nepal ranks sixth among the eight SAARC member states assessed in the 2024 E-Government Survey, ahead only of Pakistan and Afghanistan. The Maldives leads the region with a global rank of 94 and a score of 0.6745, followed by India at rank 97 (score 0.6678), Sri Lanka at rank 98 (score 0.6667), Bangladesh at rank 100 (score 0.6570), and Bhutan at rank 103 (score 0.6511). Nepal follows at rank 119 with a score of 0.5781, ahead of Pakistan at rank 136 (score 0.5096) and Afghanistan at rank 188 (score 0.2083). The South Asian regional average EGDI in 2024 stood at 0.5855, marginally above Nepal's individual score, indicating that Nepal continues to perform slightly below the regional mean despite its recent improvement.

5.2 Strengths and Structural Deficits

A detailed sub index comparison reveals a distinctive regional pattern: Nepal's Telecommunication Infrastructure Index of 0.7653 is in fact the highest recorded among all South Asian states in the 2024 survey, exceeding Bhutan (0.8169 is Bhutan's actual TII, marginally higher than Nepal), Sri Lanka (0.7936), the Maldives (0.7886), and India (0.5700), positioning Nepal as a genuine regional leader in raw telecommunications connectivity relative to its income classification. However, Nepal's Online Service Index of 0.4481 is the lowest among all South Asian states except Afghanistan, trailing India (0.8184), Bangladesh (0.7374), Pakistan (0.7042), Bhutan (0.5886), the Maldives (0.6220), and Sri Lanka (0.5494) by a wide margin. This finding decisively confirms that Nepal's comparative regional disadvantage lies not in physical network access, where the country is in fact a regional leader, but specifically in the transactional maturity, institutional framework, and content depth of its online government services, a gap most plausibly attributable to slower digitization of core administrative workflows, weaker cross agency data integration, and comparatively limited investment in citizen facing application development relative to network infrastructure.

5.3 Landlocked Developing Country (LLDC) Context

The 2024 UN E-Government Survey explicitly identifies landlocked developing country status, a classification shared by Nepal, Bhutan, Rwanda, and Zambia among others, as a structural factor complicating digital government expansion, owing to elevated costs of international bandwidth procurement that must transit neighboring coastal or better connected states. For Nepal, international internet bandwidth is procured predominantly through Indian telecommunications gateways at Birgunj, Bhairahawa, and Biratnagar, with more recent diversification through Chinese cross border fiber links via the Rasuwagadhi corridor following the 2018 Nepal China cross border optical fiber agreement. This dependency structure introduces both cost and geopolitical resilience considerations that do not affect maritime access South Asian states such as Sri Lanka, Bangladesh, or the Maldives to the same degree, and represents a structural, geography driven constraint that domestic policy alone cannot fully resolve, though bandwidth cost pressures have moderated considerably as competitive transit arrangements and cross border fiber diversification have matured over the past decade.

Table 4: South Asian Regional Comparison Matrix (EGDI, OSI, TII, HCI, and EPI Metrics, 2024)

Country	Global Rank	EGDI	OSI	TII	HCI	EPI
Maldives	94	0.6745	0.6220	0.7886	0.6130	0.4795
India	97	0.6678	0.8184	0.5700	0.6149	0.6575
Sri Lanka	98	0.6667	0.5494	0.7936	0.6570	0.4110
Bangladesh	100	0.6570	0.7374	0.6501	0.5834	0.6164
Bhutan	103	0.6511	0.5886	0.8169	0.5478	0.4932
Nepal	119	0.5781	0.4481	0.7653	0.5210	0.2192
Pakistan	136	0.5096	0.7042	0.4745	0.3500	0.4932
Afghanistan	188	0.2083	0.1438	0.2167	0.2643	0.1096

Table 5: Telecommunication and Affordability Benchmark Across South Asia

Country	TII Rank Within Region	Relative Connectivity Position	Principal International Bandwidth Route
Nepal	1st	Regional leader in raw telecommunications infrastructure despite lower middle income status	Indian gateways (Birgunj, Bhairahawa, Biratnagar) and China cross border fiber via Rasuwagadhi

Country	TII Rank Within Region	Relative Connectivity Position	Principal International Bandwidth Route
Bhutan	2nd	High connectivity supported by concentrated population and terrain specific microwave and fiber links	India gateway via Phuentsholing
Sri Lanka	3rd	Strong connectivity supported by submarine cable landing stations	Direct submarine cable access (SEA ME WE system)
Maldives	4th	High connectivity across dispersed island geography via satellite and submarine cable	Submarine cable and satellite backhaul
Bangladesh	5th	Strong connectivity supported by submarine cable and expanding terrestrial fiber	Submarine cable landing at Cox's Bazar
India	6th	Large absolute infrastructure base with substantial urban rural disparity	Multiple submarine cable landing stations
Pakistan	7th	Moderate connectivity constrained by security related service disruptions in some regions	Submarine cable landing at Karachi
Afghanistan	8th	Severely constrained connectivity amid infrastructure damage and instability	Regional transit via Pakistan, Iran, and Central Asian states

CHAPTER 6: DOMESTIC POLICY, INFRASTRUCTURE, AND DIGITAL SERVICE AUDIT

6.1 Strategic Blueprint Review: The Digital Nepal Framework (DNF) 2019

The Digital Nepal Framework, launched in 2019 by the Ministry of Communications and Information Technology, remains Nepal's principal national digital transformation strategy document. The framework is organized around eight priority sectors, namely Digital Foundation, Agriculture, Health, Education, Energy, Tourism, Finance, and Urban Infrastructure, and it identifies eighty specific digital initiatives intended to be implemented across a phased timeline extending toward 2025. The Digital Foundation pillar, which is of the most direct relevance to EGDI performance, encompasses the National Optical Fiber Backbone, the Government Integrated Data Center at Singha Durbar, the disaster recovery center at Hetauda, national identity infrastructure, and the establishment of the Nepal Government Enterprise Architecture. While a substantial share of Digital Foundation initiatives (particularly those relating to physical network infrastructure and data center capacity) have progressed to advanced implementation stages, a comparatively smaller share of transactional service and interoperability initiatives across the remaining seven sectoral pillars have achieved full operational status, mirroring the OSI TII asymmetry identified in Chapter 4.

6.2 Data Center and Network Backbone

The Government Integrated Data Center (GIDC) at Singha Durbar functions as the primary sovereign hosting facility for federal government digital platforms, hosting core databases, application servers, and network infrastructure for major citizen facing systems. The Hetauda Disaster Recovery Center provides a geographically separated backup and business continuity facility, intended to safeguard government data and service continuity in the event of a major disruption at the primary Singha Durbar facility, a design consideration of particular importance given Nepal's high seismic risk profile. The National Optical Fiber Backbone, substantially financed through the Rural Telecommunications Development Fund (RTDF), has progressively extended fiber connectivity to district headquarters and an increasing number of local government

units, forming the physical substrate upon which the dramatic TII gains documented in Chapter 4 have been achieved. Notwithstanding these advances, the effective utilization of this backbone for last mile connectivity to individual government offices, health posts, and schools in remote hill and mountain districts remains incomplete, with connectivity quality and reliability continuing to vary substantially by geographic region.

6.3 Flagship Citizen Systems Audit

Nepal has deployed a substantial portfolio of citizen facing digital platforms over the past decade, of which the following represent the most significant in terms of population reach and administrative importance. The Nagarik App, developed and maintained under the coordination of the National Information Technology Center, functions as a unified mobile gateway intended to consolidate access to multiple government services including tax payment, vital registration certificate requests, and various local government services, though the depth of true end to end transactional completion available within the app varies considerably by service category. The National Identity Card system, administered by the Department of National ID and Civil Registration (DoNIDCR), has issued biometric national identity cards to a majority of the eligible population since its phased rollout began in 2020, and increasingly functions as the foundational identity layer underpinning other digital government services, though full nationwide coverage and integration with all dependent systems remains an ongoing process. The e-Passport system has transitioned Nepal's passport issuance process to biometric, machine readable format, reducing processing timelines relative to the prior manual system, though demand driven backlogs persist during peak application periods, particularly associated with the surges in foreign labor migration. The Department of Transport Management's Electronic Driving License and Vehicle Registration System (EDLRS) has digitized substantial portions of the driving license and vehicle registration workflow, but continues to experience well documented service backlogs and appointment scheduling bottlenecks, particularly in the Kathmandu Valley, reflecting capacity constraints relative to demand rather than purely technological limitations. The e-Government Procurement (e-GP) system, operated by the Public Procurement Monitoring Office, has substantially digitized public tender publication and bid submission processes, improving procurement transparency, though integration between e-GP and downstream financial management and audit systems remains only partially realized. The Inland Revenue Department's Integrated Tax System has

enabled online tax registration, filing, and payment for a growing share of taxpayers, representing one of the more transactionally mature platforms within the government's digital portfolio. The Land Records Information Management System (LRIMS) has digitized land ownership and transaction records in a growing number of districts, though nationwide digitization and integration with municipal level land administration offices remains incomplete. Finally, the SUTRA municipal finance system has been adopted by a majority of local governments for budget planning, expenditure tracking, and financial reporting, representing a significant, though still evolving, achievement in subnational public financial management digitization.

6.4 Institutional and Federal Friction

Nepal's transition to a federal governance structure under the 2015 Constitution has introduced significant institutional complexity into digital government coordination. Data silos persist among federal ministries, each of which has historically procured and maintained independent information systems with limited standardized interoperability, a pattern reinforced by the absence, until relatively recently, of a binding, government wide enterprise architecture mandate. Vendor lock in represents a further structural constraint, as numerous government systems have been developed under bespoke contracts with limited source code ownership or transferability provisions, complicating subsequent system integration, maintenance transitions, and long term total cost of ownership management. At the local level, municipal digital fragmentation is pronounced, with the seven hundred and fifty three local governments exhibiting highly uneven digital capacity, ranging from municipalities with relatively mature digital service portals to rural municipalities with minimal digital infrastructure beyond basic SUTRA financial reporting compliance.

Table 6: Operational Status, Technology Stacks, and Transactional Depth of Core Nepali Citizen Portals

Platform	Administering Agency	Primary Function	Transactional Depth	Interoperability Status
Nagarik App	National Information Technology Center (NITC)	Unified citizen services gateway	Partial (varies by integrated service)	Moderate, expanding
National ID Card	DoNIDCR	Biometric national identity	Foundational identity layer	Increasing, foundational for other systems

Platform	Administering Agency	Primary Function	Transactional Depth	Interoperability Status
e-Passport	Department of Passports	Biometric passport issuance	High for issuance workflow	Limited external integration
EDLRS	Department of Transport Management (DoTM)	Driving license and vehicle registration	Moderate, backlog constrained	Limited
e-GP	Public Procurement Monitoring Office (PPMO)	Public tender and bid management	High for tender publication and bidding	Partial integration with financial systems
Integrated Tax System	Inland Revenue Department (IRD)	Tax registration, filing, and payment	High	Moderate
LRIMS	Department of Land Management	Land record digitization	Moderate, district variable	Limited, expanding
SUTRA	Ministry of Federal Affairs and General Administration	Municipal financial management	High for budget and expenditure tracking	Moderate across local governments

Table 7: Alignment Analysis Between DNF 2019 Initiatives and UN EGDI Dimensions

DNF 2019 Sectoral Pillar	Primary EGDI Dimension Addressed	Implementation Maturity
Digital Foundation	TII (primarily); partial HCI	Advanced (infrastructure and data centers largely operational)
Agriculture	OSI (service provision)	Early to moderate
Health	OSI (service provision); EPI (information)	Moderate, accelerated post pandemic
Education	HCI (indirect); OSI	Moderate
Energy	OSI (institutional and service provision)	Early
Tourism	OSI (content and service provision)	Moderate
Finance	OSI (transactional depth)	Moderate to advanced (tax and procurement systems)
Urban Infrastructure	OSI; EPI (municipal consultation)	Early to moderate, highly variable by municipality

CHAPTER 7: STRUCTURAL ANALYSIS AND IMPLEMENTATION CHALLENGES

7.1 The OSI TII Divergence: Infrastructure Rich, Service Poor Governance

The central structural finding of this report, introduced in Chapter 4 and confirmed through regional benchmarking in Chapter 5, is that Nepal exhibits a pronounced condition that may be described as infrastructure rich, service poor governance. This condition arises when substantial capital investment and policy attention are directed toward network infrastructure, a domain in which measurable progress is comparatively straightforward to achieve through procurement of towers, fiber, and hardware, while the more institutionally demanding work of redesigning administrative workflows, digitizing back office processes, training civil servants, and building citizen facing transactional interfaces receives comparatively less sustained investment and political priority. This pattern is not unique to Nepal among landlocked and lower middle income states, but the magnitude of the divergence observed in Nepal's 2024 sub index scores, a TII more than seventy percentage points higher than its OSI, is among the most pronounced documented within the South Asian region.

7.2 Systemic Reliability and Downtime

Multiple Nepali government digital platforms, including the Integrated Tax System during annual tax filing deadlines and the EDLRS during periods of concentrated driving license examination scheduling, have experienced well documented concurrency failures and server bottlenecks during periods of high simultaneous user demand. These reliability failures reflect a combination of insufficient server capacity provisioning relative to peak demand, limited investment in horizontally scalable cloud native application architectures, and an absence of dedicated performance engineering and load testing practices within the government software development lifecycle, rather than fundamental deficiencies in the underlying network infrastructure itself.

7.3 Data Fragmentation and Interoperability Gaps

Resistance to full adoption of the Nepal Government Enterprise Architecture (NGEA) and the Nepal Government Interoperability Framework (NeGIF) persists across numerous federal ministries and agencies, many of which continue to operate legacy systems developed under independent, siloed procurement processes predating these frameworks. The absence of a mandatory, centrally enforced application programming interface standard has meant that even where individual systems, such as the National ID system and the Integrated Tax System, contain overlapping citizen data, automated cross system data exchange remains the exception rather than the norm, requiring citizens to repeatedly submit the same documentary evidence across multiple agencies for related administrative purposes.

7.4 Cybersecurity Posture and Human Capital Drain

Government domains under the dot gov dot np namespace have experienced periodic distributed denial of service (DDoS) attacks and website defacement incidents in recent years, incidents that have exposed the absence of a centralized, adequately resourced Security Operations Center (SOC) capable of providing continuous monitoring and rapid incident response across the full portfolio of government digital assets. Compounding this vulnerability, Nepal's civil service technical cadres continue to operate under compensation structures substantially below private sector and international market rates for comparable software engineering, systems administration, and cybersecurity skill sets, contributing to a persistent brain drain of trained ICT professionals toward private sector employment or international labor migration, further constraining the government's institutional capacity to maintain, secure, and modernize its expanding digital infrastructure portfolio.

Table 8: Structural Barriers to EGDI Elevation Across Policy, Technology, and Human Capital Dimensions

Dimension	Principal Structural Barrier	Affected EGDI Sub Index
Policy	Incomplete enforcement of NGEA and NeGIF interoperability mandates	OSI
Policy	Pending status of the IT and Cyber Security Bill	OSI; institutional framework

Dimension	Principal Structural Barrier	Affected EGDI Sub Index
Technology	Legacy, siloed agency systems lacking API based integration	OSI
Technology	Insufficient scalable cloud architecture for peak demand periods	OSI
Human Capital	Civil service wage compression driving ICT talent attrition	HCI (institutional capacity); OSI
Human Capital	Digital literacy gap between general and government specific ICT competency	HCI
Institutional	Persistent data silos among federal, provincial, and local governments	OSI; EPI
Institutional	Absence of institutionalized citizen consultation mechanisms	EPI

CHAPTER 8: RISKS, CYBERSECURITY, AND OPERATIONAL THREATS

8.1 National Cyber Threat Matrix

National digital registries and public platforms holding sensitive citizen data, including the National ID system, the Integrated Tax System, and land records databases, represent high value targets for a range of threat actors, including opportunistic criminal groups seeking financial data or personally identifiable information for fraud purposes, and, at a lower probability but higher impact tier, state affiliated or politically motivated actors seeking to disrupt government service availability or deface public facing platforms for reputational or symbolic effect.

8.2 National CERT and Incident Response Deficits

While Nepal maintains a national Computer Emergency Response Team (CERT) function, its resourcing and operational scope relative to the scale of the government's expanding digital asset portfolio remains limited compared to more mature regional CERT capacities such as those operated in India or Sri Lanka. Patch management practices across government systems are inconsistently applied, and maintenance debt, meaning the accumulated backlog of unaddressed software updates, security patches, and infrastructure modernization needs, continues to accumulate across a portfolio of systems that has expanded faster than the corresponding growth in dedicated maintenance and security budget allocations.

8.3 Data Protection and Sovereignty

Nepal currently lacks a dedicated, independently empowered data privacy commissioner or equivalent regulatory authority responsible for enforcing citizen data protection standards across government and private sector data controllers, a gap that becomes increasingly consequential as the scope and sensitivity of centrally held citizen data continues to expand through systems such as the National ID database. Additionally, the procurement of third party software and cloud services without adequate security vetting protocols introduces supply chain risk, particularly

where proprietary systems are hosted outside Nepal's sovereign data center infrastructure or where source code and data ownership terms are insufficiently specified within procurement contracts.

Table 9: Risk Categorization, Severity Level, and Operational Mitigation Framework

Risk Category	Description	Severity	Recommended Mitigation
Denial of Service Attacks	Disruption of government website and portal availability	High	Establish centralized SOC with continuous monitoring and DDoS mitigation capacity
Data Breach	Unauthorized access to citizen personal data held in national registries	Critical	Establish independent data protection authority and mandatory encryption standards
Legacy System Vulnerability	Unpatched or outdated software exposing exploitable vulnerabilities	High	Institute mandatory patch management schedule and system modernization roadmap
Third Party Supply Chain Risk	Unvetted vendor software introducing hidden vulnerabilities	Medium	Mandatory security audit and source code escrow provisions in procurement contracts
Insider Threat and Human Error	Misconfiguration or unauthorized data access by government personnel	Medium	Role based access control and mandatory cybersecurity training for civil servants
Natural Disaster Disruption	Seismic or flood related damage to hosting infrastructure	Medium	Continued investment in the Hetauda Disaster Recovery Center and geographic redundancy

CHAPTER 9: STRATEGIC ROADMAP FOR ADVANCING TO VERY HIGH EGDI (VHEGDI)

Advancing from the High EGDI category (scores between 0.50 and 0.75) to the Very High EGDI category (scores exceeding 0.75) will require Nepal to address the structural OSI TII asymmetry identified throughout this report as its central strategic priority. Given that Nepal's TII already exceeds 0.75, further composite score gains will depend disproportionately upon closing the gap in OSI and, to a lesser extent, HCI performance, since continued TII investment alone offers diminishing marginal returns to the composite EGDI score once that sub index approaches saturation.

Core Prerequisites for VHEGDI Transition

1. Full nationwide enforcement of the Nepal Government Enterprise Architecture and Nepal Government Interoperability Framework across all federal ministries, provincial governments, and local governments, with binding compliance timelines and penalties for continued non integration.
2. Completion of end to end transactional digitization for the highest volume citizen services, extending beyond current partial implementations toward full online completion without any requirement for in person follow up.
3. Institutionalization of a permanent, adequately funded e-consultation mechanism integrated into the legislative and budgetary process at federal, provincial, and local levels, addressing the structural EPI weakness identified in Chapter 4.
4. Establishment of a dedicated, competitively compensated government technology cadre with career progression pathways designed to reduce ICT talent attrition toward the private sector.

Expanding API Orchestration, Sovereign Cloud Infrastructure, and Open Government Data

A national API gateway, mandatorily adopted across all government information systems, would allow the substantial existing data holdings within the National ID system, the Integrated Tax System, and land records databases to be securely exchanged between authorized government agencies without requiring citizens to resubmit identical documentation. Expansion of sovereign cloud infrastructure, building upon the existing Government Integrated Data Center and Hetauda Disaster Recovery Center, would provide the elastic compute capacity required to eliminate the concurrency failures documented in Chapter 7. A national Open Government Data (OGD) portal, publishing machine readable datasets across budget, procurement, and service delivery domains, would directly strengthen Nepal's OSI content provision and e-information sub dimensions.

Institutionalizing Artificial Intelligence in Public Service Delivery

Targeted deployment of artificial intelligence capabilities within specific high volume administrative functions, including automated triage and routing of public grievance redressal complaints, predictive maintenance scheduling for municipal infrastructure, and anomaly detection for tax fraud identification within the Integrated Tax System, represents a comparatively lower cost pathway to service quality improvement relative to full scale system replacement, and aligns with the special artificial intelligence supplement featured in the 2024 UN E-Government Survey, which identifies AI adoption as an emerging differentiator among Very High EGDI states.

CHAPTER 10: CONCLUSION AND STRATEGIC SUMMARY

This case study has traced Nepal's two decade long trajectory within the United Nations E-Government Development Index, documenting the country's historic entry into the High EGDI rating class in the 2024 survey cycle with a composite score of 0.5781 and a global rank of 119 out of 193 member states. The analysis has identified a fundamental structural asymmetry underlying this achievement: Nepal's telecommunications infrastructure, as measured by the Telecommunication Infrastructure Index, has advanced at a pace disproportionate to the country's transactional online service maturity and citizen participation infrastructure, as measured respectively by the Online Service Index and the supplementary E-Participation Index.

Regional benchmarking against South Asian peers confirms that Nepal's comparative disadvantage is concentrated specifically in online service depth and citizen participation mechanisms rather than in physical network connectivity, where Nepal in fact leads the region. Domestic policy analysis further reveals that while the Digital Nepal Framework 2019 has succeeded in advancing the Digital Foundation pillar, corresponding progress across the framework's remaining seven sectoral pillars, and the institutional interoperability infrastructure required to connect them, has lagged.

The unifying strategic imperative that emerges from this analysis is the necessity of a deliberate policy and investment reallocation away from continued incremental telecommunications infrastructure expansion, a domain in which Nepal has already achieved regional leadership and diminishing marginal returns are increasingly likely, toward sustained, institutionally difficult investment in transactional service digitization, cross agency interoperability, citizen participation infrastructure, and human capital development within the government's own technical workforce. Without this reallocation, Nepal's EGDI trajectory risks plateauing within the High EGDI category, unable to accumulate the additional OSI and EPI gains required to cross the 0.75 threshold into the Very High EGDI category occupied by regional aspirational benchmarks and global leaders alike.

CHAPTER 11: ACTIONABLE RECOMMENDATIONS

11.1 Federal Policy Makers and MoCIT

1. Expedite the passage of the pending IT and Cyber Security Bill to establish a comprehensive, modern statutory foundation for digital governance, data protection, and cybersecurity enforcement, superseding the increasingly dated Electronic Transactions Act 2063.
2. Mandate binding, time bound compliance deadlines for full agency adoption of the Nepal Government Enterprise Architecture and Nepal Government Interoperability Framework, with clear accountability mechanisms for non compliant ministries.
3. Establish an independent data protection authority empowered to enforce citizen data privacy standards across both government and private sector data controllers.

11.2 Infrastructure and Technical Operators

1. Direct the National Information Technology Center, in coordination with the Nepal Telecommunications Authority, to modernize the Government Integrated Data Center toward an elastic, horizontally scalable cloud architecture capable of absorbing peak demand surges without service degradation.
2. Establish a centrally resourced, twenty four hour Security Operations Center with continuous monitoring, DDoS mitigation capacity, and rapid incident response protocols across the full government digital asset portfolio.

11.3 Implementing Ministries and Agencies

1. Prioritize application programming interface based integration between the Department of Transport Management, the Department of National ID and Civil Registration, the Inland Revenue Department, and the Public Procurement Monitoring Office to eliminate redundant citizen documentation submission requirements.

2. Institute mandatory load testing and performance engineering practices prior to the launch of any new citizen facing transactional platform, informed by lessons from documented EDLRS and Integrated Tax System capacity failures.

11.4 Local and Provincial Governments

1. Standardize municipal digital service platforms through a common, centrally maintained technology template extending the SUTRA model beyond financial management into broader citizen service delivery, reducing the current wide variance in local government digital maturity.
2. Establish institutionalized, recurring digital consultation mechanisms at the municipal level, directly addressing the structural E-Participation Index weakness identified throughout this report.

11.5 Academia and Domestic ICT Industry

1. Reform computer science and information technology curricula across Nepali universities, including Tribhuvan University affiliated colleges, to incorporate dedicated coursework in government enterprise architecture, public sector systems interoperability, and digital governance policy alongside conventional software engineering training.
2. Expand structured public private partnership models between domestic ICT firms and government implementing agencies to supplement civil service technical capacity constraints while retaining source code ownership and long term maintainability within government control.

Table 10: Prioritized National EGDI Elevation Action Plan with Phased Horizons

Action	Responsible Stakeholder	Time Horizon
Passage of the IT and Cyber Security Bill	Federal Parliament; MoCIT	Short Term (within 1 year)
Establishment of centralized Security Operations Center	NITC; Nepal Telecommunications Authority	Short Term (within 1 year)
Mandatory NGEA and NeGIF compliance deadlines for all ministries	MoCIT; Office of the Prime Minister and Council of Ministers	Medium Term (1 to 3 years)

Action	Responsible Stakeholder	Time Horizon
Full API integration among DoTM, DoNIDCR, IRD, and PPMO	Respective implementing agencies	Medium Term (1 to 3 years)
Establishment of independent data protection authority	Federal Parliament; Ministry of Law, Justice and Parliamentary Affairs	Medium Term (1 to 3 years)
Institutionalized e-consultation infrastructure at all governance tiers	MoCIT; Ministry of Federal Affairs and General Administration; local governments	Long Term (3 to 5 years)
Dedicated, competitively compensated government technology cadre	Public Service Commission; Ministry of Finance	Long Term (3 to 5 years)
Curriculum reform across CSIT and ICT academic programs	Tribhuvan University; affiliated colleges; University Grants Commission	Long Term (3 to 5 years)

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